

END TERM EXAMINATION
THIRD SEMESTER [B.TECH] DECEMBER 2016

Paper Code: ETIT-209**Subject: Object Oriented Programming using C++**

Time: 3 Hours
Maximum Marks: 75

Note: Attempt any five questions including Q. No. 1 which is compulsory.

Q. No.	Question	Marks
1	Attempt the following questions: (a) What kind of things can become objects in OOPS? (b) Define private, public and protected data members of a class with examples. (c) What do you mean by dynamic binding? How is it useful in C++? (d) Distinguish between data abstraction and data encapsulation. (e) Describe with examples, the uses of enumeration data types in C++. (f) Differentiate between constructor and destructor. (g) How do the following statements differ? (i) <code>char * const p;</code> (ii) <code>char const * p;</code> (h) When do we declare any member of a class as static? (i) What is a parameterized constructor? Explain using an example. (j) Define operators which cannot be overloaded in C++.	10 × 2.5 = 25
2	(a) What is the object-oriented paradigm? Explain the various features of object-oriented paradigm. (8.5 Marks) (b) What is the difference between member functions defined inside and outside the body of a class? (4 Marks)	8.5 + 4 = 12.5
3	(a) Define operator overloading. Write a program in C++ to subtract two complex numbers using operator overloading. (8 Marks) (b) Explain how base class member function can be invoked in a derived class if the derived class also has a member function with the same name. (4.5 Marks)	8 + 4.5 = 12.5

[P.T.O.]

Q. No.	Question	Marks
4	(a) What are friend functions and friend classes? Discuss the relation of friend functions with public and protected data members of the class. (6 Marks) (b) Overload new and delete operators to dynamically allocate and manipulate objects of any assumed class. (6.5 Marks)	6 + 6.5 = 12.5
5	(a) Define the following terms with suitable illustrations: (3 × 2.5 = 7.5 Marks) (i) Destructor (ii) Abstract Class (iii) Virtual Function (b) What is inheritance? What are the different visibility modes supported by C++? Explain with example. (5 Marks)	7.5 + 5 = 12.5
6	(a) What is a template function in C++? Write a program for template function overloading. (6.5 Marks) (b) Write a program that reads content from one file and copies that content to another file using C++ file streams. (6 Marks)	6.5 + 6 = 12.5
7	(a) What are exceptions? What are the differences between synchronous and asynchronous exceptions? (6.5 Marks) (b) What is the Standard Template Library (STL)? Explain the terms sequence containers and associative containers with examples. (6 Marks)	6.5 + 6 = 12.5
